CRT42 G2 Trouble Shooting And FAQ List

Contents

Tools： Computer、CAN DEBUG TOOL（ERP: R0203845-00）、16-Pin adapter（ERP: R0203075-00）、Jack、Multimeter.

CAN DEBUG TOOL（ERP: R0203845-00）| 16-Pin adapter（ERP: R0203075-00）

Jack | Multimeter

PMU2 - Overtemperature-Level1

Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, replace the battery compartment.

PMU10 - PMU loop fault

At least one channel in the battery compartment is unable to discharge, but the performance will not decrease, vehicle still can be worked.

If you feel a decrease in mowing performance with this fault code, the battery compartment should be replaced.

PMU11 - PMU minor fault

There have occurred a over-current(Level 1) or over-temperature(Level 1) fault. The vehicle's performance will be limited with this code, but this fault can be restored, no need to pay attention.

PMU12 - PMU major fault

Check：Use "ToolsForCAN" to identify the detail fault -

1. "PreCharge Function Error" of PMU12

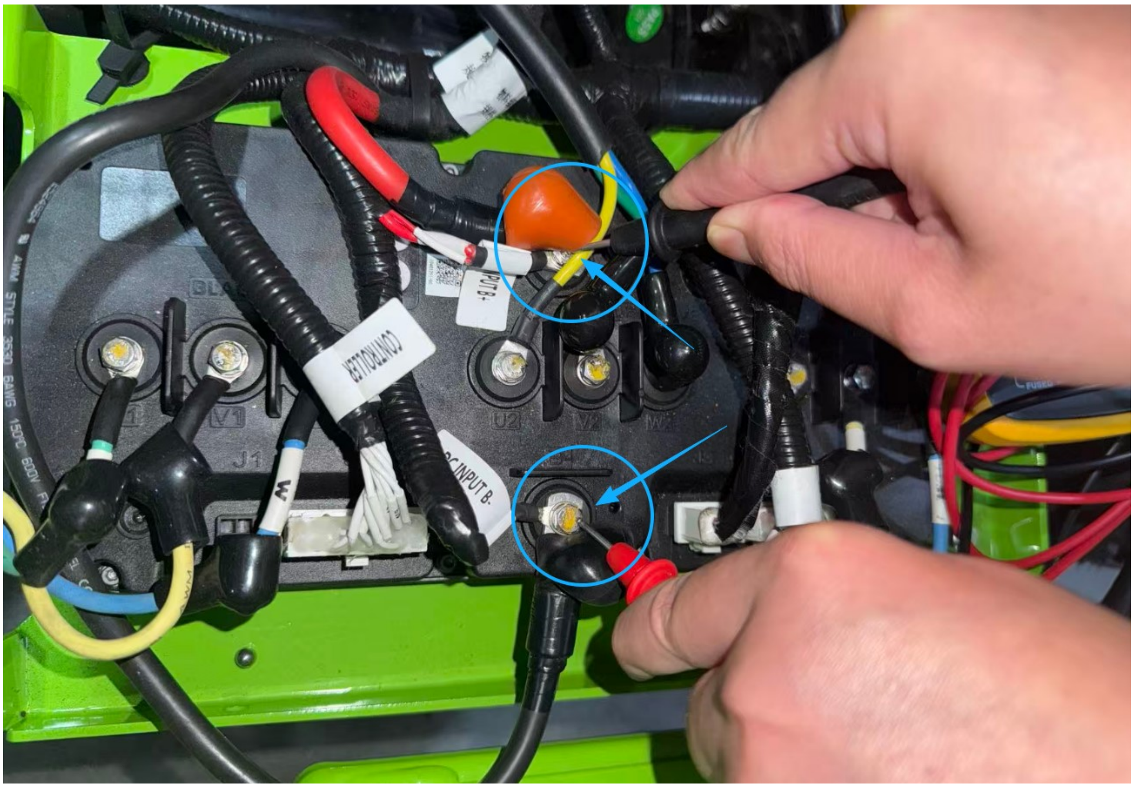
PMU FAULT LIST PART1

Function	Status	
Avg Curr Overcurr	◉ 0：Normal ○ 2：Avg Curr L2	○ 1：Avg Curr L1 ○ 3：Spare
PMUNeg. MOSOvertemp	◉ 0：Normal ○ 2：OvertempL2	○ 1：OvertempL1 ○ 3：NTCError
Peak Curr Overcurr	◉ 0：Normal	○ 1：Error
Feedback Curr Overcurr	◉ 0：Normal	○ 1：Error
PreCharge Function	○ 0：Normal	◉ 1：Error
PMUSoftware Auth	◉ 0：Normal	○ 1：Error
PMU Undervoltage Error	◉ 0：Normal	○ 1：Error
PMU Overvoltage Error	◉ 0：Normal	○ 1：Error

① Adjust the dial position of a multimeter to the diode position to measure the diode value of the controller's B+ and B- as shown in the picture below(Red lead to black wire(B-), black lead to red wire(B+)).

a. If the value is approximately equal to 0.7±0.2V， you should replace a new battery compartment.

b. Otherwise follow the steps.

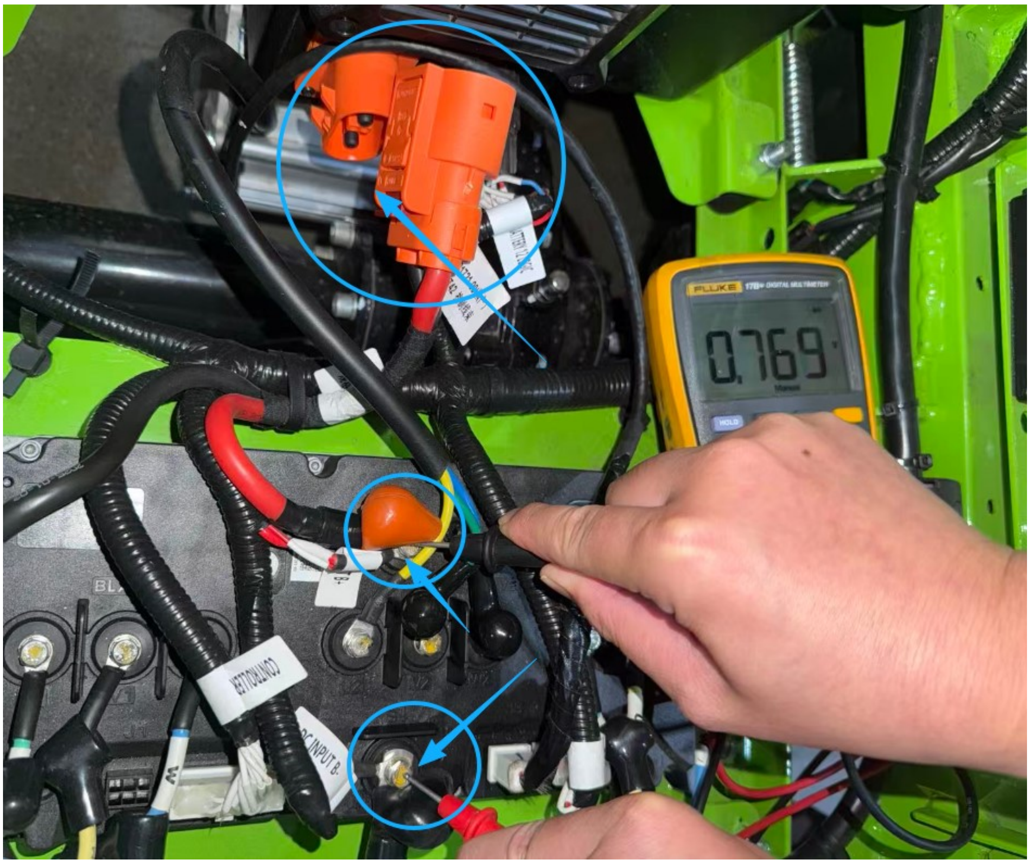


Correct value should be approximately equal to 0.7±0.2V

② Disconnect the power harness that connect to the battery box, and use a multimeter to measure the diode value as shown in the picture below(Red lead to negative electrode, black lead to positive electrode).

c. If the value is approximately equal to 0.7±0.2V, you should replace a new battery compartment.

a. Otherwise follow the steps.

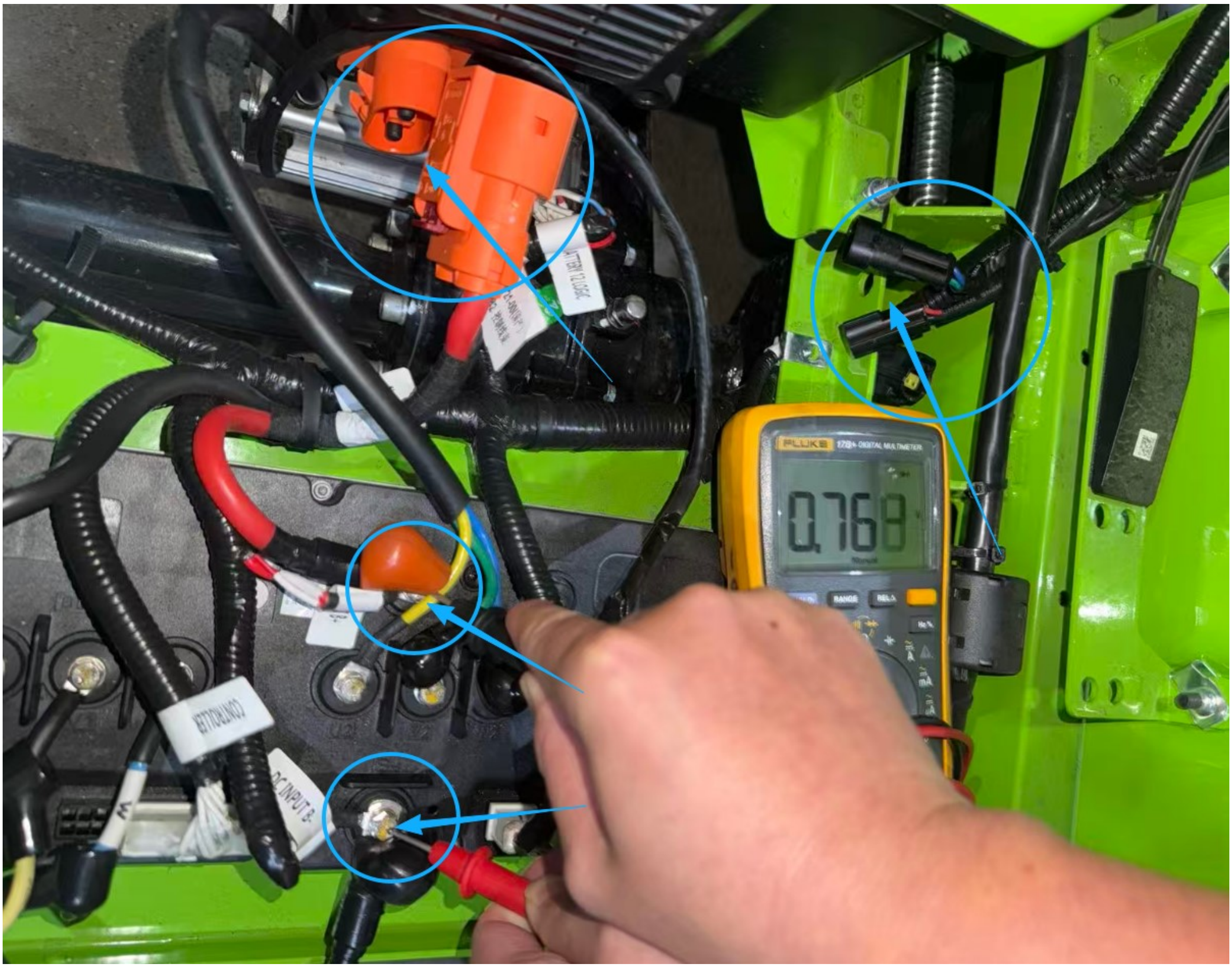


Correct value should be approximately equal to 0.7±0.2V

③ Disconnect the connectors of DC-DC and use a multimeter to measure the diode value of the controller as the picture showed below(Red lead to negative electrode, black lead to positive electrode).

a. If the value is approximately equal to 0.7±0.2V， you should replace the DC-DC.

b. Otherwise replace the controller.



Correct value should be approximately equal to 0.7±0.2V

2. Other errors of PMU12

   Replace the battery compartment.

PMU13 - PMU no available battery

1. Make sure that the inserted batteries are correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU35 - Overtemperature-Level2

Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, replace the battery compartment.

PMU36 - Undervoltage

1. Charge the batteries.
2. Make sure the battery compartment on the correct platform is used.
3. Replace the battery compartment.

PMU37 - Overvoltage

1. Make sure the battery compartment on the correct platform is used.
2. Replace the battery compartment.

PMU38 - Power supply output failure

Disconnect the power harness from battery compartment:

1. If the fault code disappeared, follow the “PMU12” trouble shooting guide to find the detail issue.
2. Otherwise, replace the battery compartment.

PMU41 - No battery pack available

1. Make sure that the inserted batteries are correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU46 - Relay（MOS） error

Replace the battery compartment.

PMU47 - Pre-charge error

Follow the “PreCharge Function Error” of PMU12 trouble shooting guide to find the detail issue.

PMU48 - Pre-charge hardware error

Replace the battery compartment.

PMU49 - Negative MOSFET temperature sensor error

Replace the battery compartment.

PMU52 - Current sensor error

Replace the battery compartment.

PMU57 - KSI Pre-MOS error

Replace the battery compartment.

PMU59 - KSI MOS error

Replace the battery compartment.

PMU61 - Battery pack 1 does not match

1. Make sure that the inserted battery in cabin 1 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU62 - Battery pack 2 does not match

1. Make sure that the inserted battery in cabin 2 is correct, not mixed with different voltage platform.

2. Replace the battery compartment.

PMU63 - Battery pack 3 does not match

1. Make sure that the inserted battery in cabin 3 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU64 - Battery pack 4 does not match

1. Make sure that the inserted battery in cabin 4 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU65 - Battery pack 5 does not match

1. Make sure that the inserted battery in cabin 5 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU66 - Battery pack 6 does not match

1. Make sure that the inserted battery in cabin 6 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU67 - Abnormal charging status

1. Check if the charger's input plug is properly plugged in.
2. Check if the vehicle can be powered on and worked properly:
   ① Can't be powered on and worked, then replace the battery compartment.
   ② Otherwise, replace the charge socket.

T2 - Drive motor stalled

Turn off the power, use a jack to lift the vehicle's rear wheels, disengage the parking brake, try to rotate the rear wheel.

a. If the wheel can't be rotated, replace the traction motor.

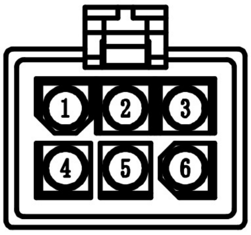
b. Otherwise, replace the controller.

T4 - Motor overspeed protection

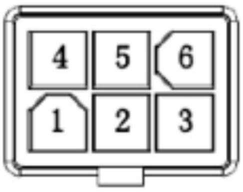
1. Replace the traction motor.
2. Replace the controller.

T5 - Motor encoder error

1. Check the 6 pins encoder sensor connector of traction motor, if any pins are bent, try to fix it, otherwise, replace the traction motor.



2. Check the 6 pins socket of controller(J3), if any pins are bent, try to fix it, otherwise, replace the controller.



3. Replace the traction motor.
4. Replace the controller.

T6 - Controller phase loss

1. Check the phase wires' terminals of traction motor, if broken, try to fix it.
2. Use multimeter to measure the resistance between UVW of the traction phase wires.

① Measure the resistance between UV, the correct value should be around 1Ω, if the value is infinite, that means the wires are broken between UV, need to replace a new traction motor.

② Measure the resistance between UW, the correct value should be around 1Ω, if the value is infinite, that means the wires are broken between UW, need to replace a new traction motor.

③ Measure the resistance between VW, the correct value should be around 1Ω, if the value is infinite, that means the wires are broken between VW, need to replace a new traction motor.

3. Replace the controller.

T7 - MOSFET error

Replace the controller.

T8 - Controller undervoltage

1. Charge the batteries.
2. Make sure the controller on the correct platform is used.
3. Flash the right firmware to the controller. 60V and 80V platform have different firmware.
4. Replace the controller.

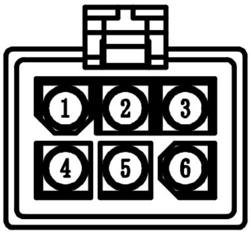
T9 - Controller overvoltage

1. Make sure the controller on the correct platform is used.
2. Flash the right firmware to the controller. 60V and 80V platform have different firmware.
3. Replace the controller.

T12 - Motor overtemperature

1. Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, follow the steps below.

① Use multimeter to measure the resistance between PIN1(black) and PIN3(white) of the 6 pins encoder sensor connector of traction motor, the correct value should be 500~1000Ω(depend on the environment temperature). If not, replace the traction motor.



② Replace the controller.

T13 - Controller overtemperature

Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, replace the controller.

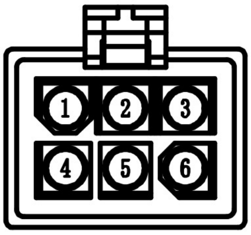
T14 - Software Overcurrent

Reduce the load, if can't solve when under normal load, try to replace a new controller.

T15 - Hardware Overcurrent

Reduce the load, if can't be solved under normal load, follow the steps below:

1. Check the 6 pins encoder sensor connector of traction motor, if any pins are bent, try to fix it, otherwise, replace the traction motor.



2. Check if all the phase wires' terminals &B+ &B- are tightened, if not, tighten them with a M5 screwdriver.
3. replace a new controller.
4. Replace a new traction motor.

T16 - Potentiometer error

Check：Use "ToolsForCAN" to read the value of potentiometer -

1. If the value is larger than 4.6 or smaller than 0.2, replace the potentiometer.
2. Replace the controller.

3. Replace the control harness.

T19 - Operating sequence error

Check：Use "ToolsForCAN" to identify the status of switches and potentiometer -

Make sure there have no operations before sitting on the seat. The status should meet the table below before operating.

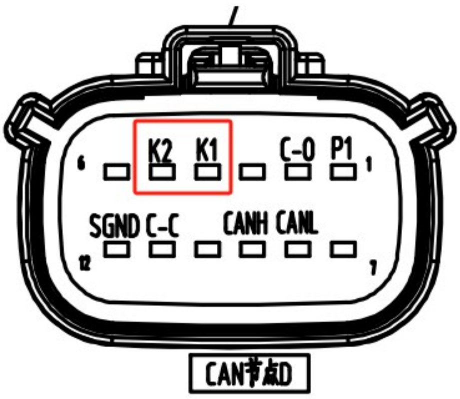
Switch & Controllers		State
Seat Switch		ON
Parking Switch		ON
PTO Switch		OFF
Cruise Button		OFF
Potentiometer	0. 80	0.6~0.95V
Controller		No Fault
PMU		No Fault

T23 - Battery compartment (PMU)communication error

1. Check the pins of connector of the battery compartment side:

a. If the pins of the controller are bent or retracted, if can be fixed, try to fix them.

b. Replace the battery compartment.



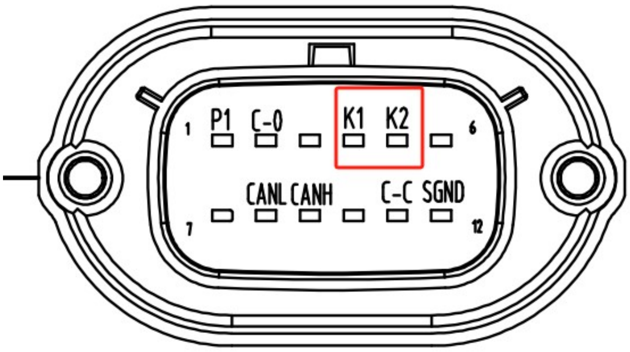
BATTERY 12 LOGIC		
供应商	名称	型号
MOLEX	外壳	334721206
	端子	330122002
	盲堵	343450001

2. Check the pins of the connector of the harness side:

a. If the pins of are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.

供应商	名称	型号
MOLEX	外壳	477259010
	端子	330000002



T24 - Battery compartment (PMU) shutdown

1. If you turn off the vehicle, you will get this fault code, it's normal.

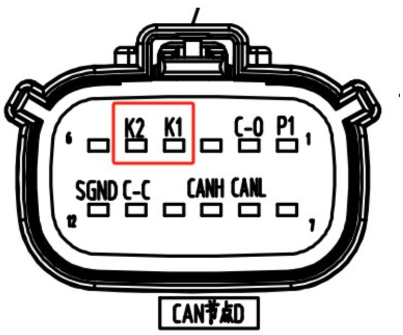
2. Otherwise:

① Check if the connector of the key switch is connected well.

② Check the pins of connector of the battery compartment side:

a. If the pins of the controller are bent or retracted, if can be fixed, try to fix them.

b. Replace the battery compartment.

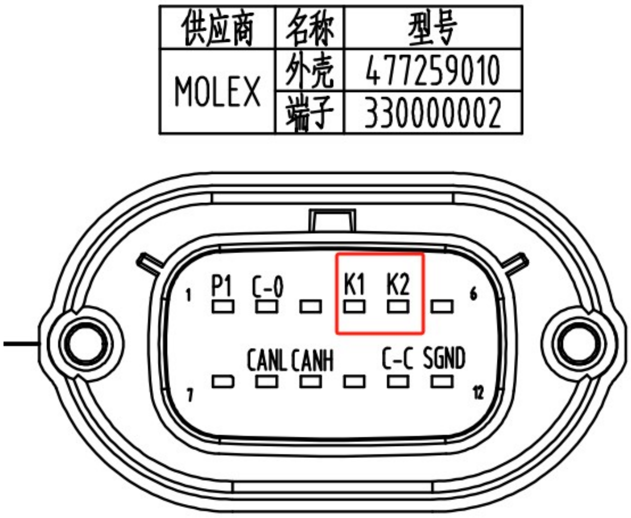


BATTERY 12 LOGIC		
供应商	名称	型号
MOLEX	外壳	334721206
	端子	330122002
	盲堵	343450001

③ Check the pins of the connector of the harness side:

a. If the pins of are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.



T27 - CAN timeout error - left blade

Flash the firmware for the left blade part first, if can't flash well, replace a new controller.

T29 - CAN timeout error - right blade

Flash the firmware for the right blade part first, if can't flash well, replace a new controller.

T31 - Seat switch verification error

1. Check if the connector of the seat switch is connected well.
2. Check the pins of connector of the controller's side:

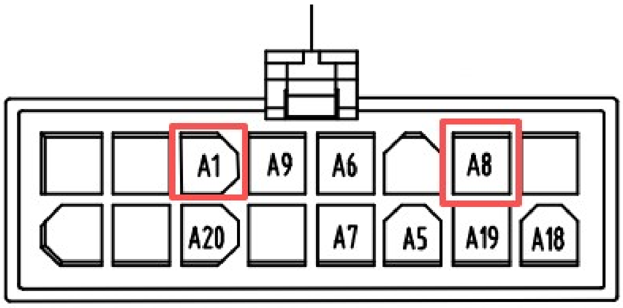
a. If the pins of the controller are bent or retracted, if can be fixed, try to fix them.

b. Replace the controller.

3. Check the pins of the connector of the harness side:

a. If the pins of are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.



CONTROLLER

供应商	名称	型号
TE	外壳	794824-1
	端子	770904-1
	单线密封	794758-1
	盲堵	794995-1
	接口密封	1-1586362-6

4. Replace the seat switch.

TR32/TL32 - Software authentication error

Replace a new controller.

MR1/ML1 - Motor positioning error

1. MR1

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have MR1 fault code, replace the controller.

b. If fault code change to ML1, you should replace the right blade motor.

2. ML1

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have ML1 fault code, replace the controller.

b. If fault code change to MR1, you should replace the left blade motor.

MR2/ML2 - Blade motor stalled

1. MR2

① Turn off the power, try to rotate the right side blade.

a. If the blade can't be rotated or very tight , replace the right blade motor.

b. Otherwise follow the steps below.

② Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

c. If still have MR2 fault code, replace the controller.

d. If fault code change to ML2, you should replace the right blade motor.

2. ML2

① Turn off the power, try to rotate the left side blade.

a. If the blade can't be rotated or very tight, replace the left blade motor.

b. Otherwise follow the steps below.

② Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

c. If still have ML2 fault code, replace the controller.

d. If fault code change to MR2, you should replace the left blade motor.

MR3/ML3 - Blade motor low speed

1. MR3

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have MR3 fault code, replace the controller.

b. If fault code change to ML3, you should replace the right blade motor.

2. ML3

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have ML3 fault code, replace the controller.

b. If fault code change to MR3, you should replace the left blade motor.

MR4/ML4 - Blade motor over speed

1. MR4

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have MR4 fault code, replace the controller.

b. If fault code change to ML4, you should replace the right blade motor.

2. ML4

Swap the M5 terminals of left and right blade motors’ phase harness(Totally six screws).

a. If still have ML4 fault code, replace the controller.

b. If fault code change to MR4, you should replace the left blade motor.

MR6/ML6 - Controller phase loss

1. MR6

Swap the M5 terminals of left and right blade motors’ phase harness(Totally six screws).

a. If still have MR6 fault code, replace the controller.

b. If fault code change to ML6, you should replace the right blade motor.

2. ML6

Swap the M5 terminals of left and right blade motors’ phase harness(Totally six screws).

a. If still have ML6 fault code, replace the controller.

b. If fault code change to MR6, you should replace the left blade motor.

MR7/ML7 - MOSFET error

Replace the controller.

MR8/ML8 - Controller undervoltage

① Charge the batteries.

② Make sure the controller on the correct platform is used.

③ Flash the right firmware to the controller. 60V and 80V platform have different firmware.

④ Replace the controller.

MR9/ML9 - Controller overvoltage

① Make sure the controller on the correct platform is used.

② Flash the right firmware to the controller. 60V and 80V platform have different firmware.

③ Replace the controller.

MR13/ML13 - Controller overtemperature

Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, replace the controller.

MR14/ML14 - Software overcurrent

1. MR14

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have MR14 fault code, replace the controller.

b. If fault code change to ML14, you should replace the right blade motor.

2. ML14

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have ML14 fault code, replace the controller.

b. If fault code change to MR14, you should replace the left blade motor.

MR15/ML15 - Hardware overcurrent

1. MR15

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have MR15 fault code, replace the controller.

b. If fault code change to ML15, you should replace the right blade motor.

2. ML15

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have ML15 fault code, replace the controller.

b. If fault code change to MR15, you should replace the left blade motor.

MR19/ML19 - Operating sequence error

1. Make sure the blade switch is on the "OFF" position, then try to turn on the blade again.
2. Check if the connector of the blade switch is connected well.

3. Check the pins of connector of the controller’s side:

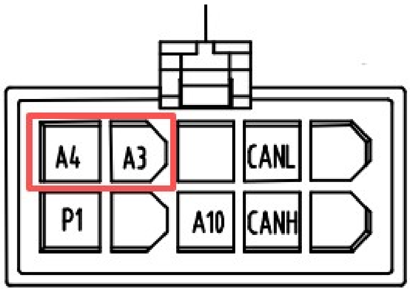
a. If the pins of the controller are bent or retracted, if can be fixed, try to fix them.

b. Replace the controller.

4. Check the pins of the connector of the harness side:

a. If the pins of are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.



CAN节点D

CONTROLLER

供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
	单线密封	794758-1
	盲堵	794995-1
	接口密封	1-794772-0

5. Replace the blade switch.

MR25/ML25 - CAN timeout error - left drive

Flash the firmware for the traction part first, if can’t flash well, replace a new controller.

MR26/ML26 - CAN timeout error - right drive

Flash the firmware for the traction part first, if can’t flash well, replace a new controller.

MR27/ML27 - CAN timeout error - left blade

Flash the firmware for the left blade part first, if can’t flash well, replace a new controller.

MR29/ML29 - CAN timeout error - right blade

Flash the firmware for the right blade part first, if can’t flash well, replace a new controller.

MR31/ML31 - Blade switch verification error

1. Check if the connector of the blade switch is connected well.
2. Check the pins of connector of the controller’s side:

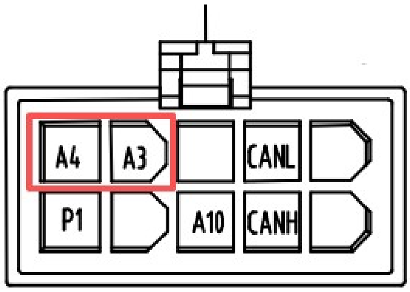
a. If the pins of the controller are bent or retracted, if can be fixed, try to fix them.

b. Replace the controller.

3. Check the pins of the connector of the harness side:

a. If the pins of are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.



CAN节点D

CONTROLLER

供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
	单线密封	794758-1
	盲堵	794995-1
	接口密封	1-794772-0

4. Replace the blade switch.

MR32/ML32 - Software authentication error

Replace the controller.

MR42/ML42 - Software version verification error

1. Flash the correct firmware for the left and right part of the controller.
2. Replace the controller.

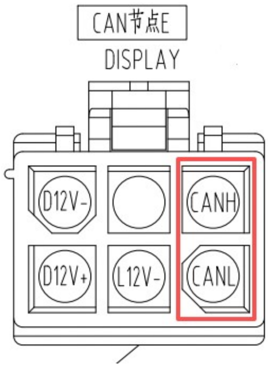
D1-CAN timeout with PMU

1. If this fault exists simultaneously with any of D2/D3/D4/D5/D6, follow the steps below.

a. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

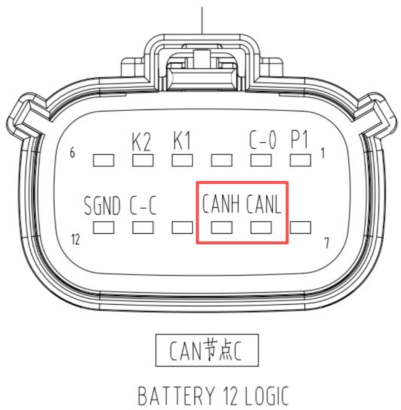
供应商	名称	型号
TE	外壳	172168-1
	端子	770988-1



2. If this fault exists alone without D2/D3/D4/D5/D6, follow the steps below.

a. Disconnect the connector of battery component side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the battery component.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.



供应商	名称	型号
MOLEX	外壳	334721206
	端子	330122002
	盲堵	343450001

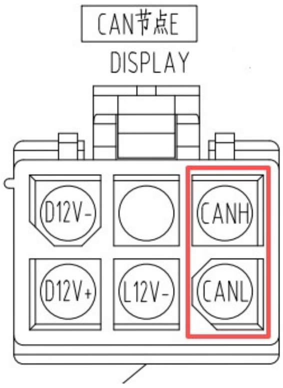
D3-CAN timeout with traction controller

1. If this fault exists simultaneously with D1, follow the steps below.

a. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

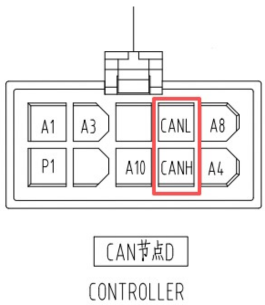
供应商	名称	型号
TE	外壳	172168-1
	端子	770988-1



2. If this fault exists without D1, follow the steps below.

a. Disconnect the connector of controller side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the controller.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.



供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
	单线密封	794758-1
	盲堵	794995-1
	接口密封	1-794772-0

D4-CAN timeout with left blade controller

1. If this fault exists simultaneously with D1, follow the steps below.

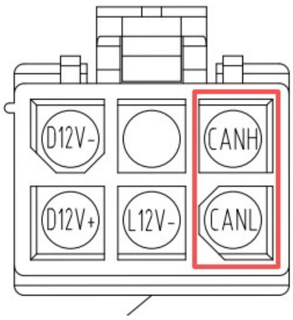
a. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

供应商	名称	型号
TE	外壳	172168-1
	端子	770988-1

CAN节点E

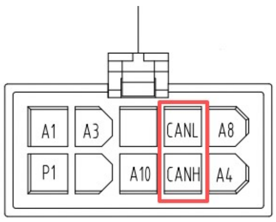
DISPLAY



1. If this fault exists without D1, follow the steps below.

a. Disconnect the connector of controller side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the controller.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.



CAN节点D

CONTROLLER

供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
	单线密封	794758-1
	盲堵	794995-1
	接口密封	1-794772-0

D6-CAN timeout with right blade controller

1. If this fault exists simultaneously with D1, follow the steps below.

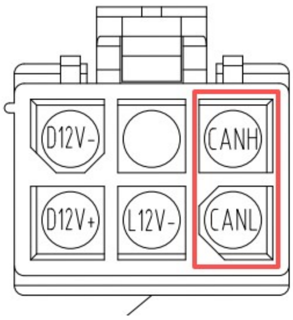
a. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

供应商	名称	型号
TE	外壳	172168-1
	端子	770988-1

CAN节点E

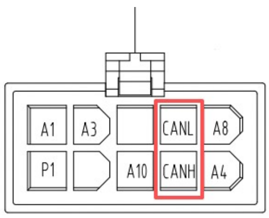
DISPLAY



2. If this fault exists without D1, follow the steps below.

a. Disconnect the connector of controller side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the controller.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.



CAN节点D

CONTROLLER

供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
	单线密封	794758-1
	盲堵	794995-1
	接口密封	1-794772-0